

Name: _____ score: _____ / 10

There are 10 points possible on this quiz. No aids (book, notes, etc.) are permitted. You may use a non-programmable calculator. **Show all work and supporting calculations for full credit. Explain how you get your answers.**

1. (3 points) Consider the weighted voting system $[31 : 16, 12, 10, 3]$

(a) What is the meaning of the number **31** in the weighted voting system?

(b) Why is $\{P_1, P_2, P_4\}$ a **winning coalition**? Explain your answer with a calculation and a few words.

(c) Which players, if any, have **veto power**? Explain your reasoning.

2. (2 points) Consider the weighted voting system $[q : 12, 8, 2, 1]$.

(a) What is the **smallest possible quota value**? Explain how you know.

(b) Determine a value of q that results in a **dictator**. Explain which player is the dictator and how you know.

3. Consider the weighted voting system $[17 : 13, 9, 6, 2]$ and the list of all possible coalitions.

coalition	weight	winning?	coalition	weight	winning?
$P_1 P_2$			$P_1 P_2 P_3$		
$P_1 P_3$			$P_1 P_2 P_4$	24	
$P_1 P_4$			$P_1 P_3 P_4$	21	
$P_2 P_3$	15		$P_2 P_3 P_4$	17	
$P_2 P_4$	11		$P_1 P_2 P_3 P_4$	30	
$P_3 P_4$	8				

- (a) (1 point) In the table above, fill in the **missing weights** of each coalition in the appropriate column.
- (b) (1 point) Put a check $\checkmark$ in the column labeled “winning?” to identify all winning coalitions, and mark $\times$ for non-winning coalitions.
- (c) (2 points) In each winning coalition, **underline the critical players**.
- (d) (1 point) Are there any **dummies**? Explain why or why not, with a sentence or short computation and sentence.